

Inflammation

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Objectives

- Know the causes of inflammation
- Understand the process of inflammations
- Comprehend the etiopathogeneses of granulomatous inflammations
- Contrast the differences between acute and chronic inflammations

Introduction

- Inflammation is a local response (reaction) of living vascularized tissues to endogenous and exogenous stimuli.
- The term is derived from the Latin "inflammare" meaning to burn.
- Inflammation is fundamentally destined to localize and eliminate the causative agent and to limit tissue injury.
- Thus, inflammation is a physiologic (protective) response to injury.

- Inflammation can be damaging if
 - Inadequately controlled
 - Misdirected
 - If stimulus is persistent cant be readily eliminatied

Causes

➤ Causes of inflammation are apparently causes of diseases such as

- Physical agents - mechanical injuries, alteration in temperatures and pressure, radiation injuries.
- Chemical agents- including the ever increasing lists of drugs and toxins.
- Biologic agents (infectious)- bacteria, viruses, fungi, parasites
- Immunologic disorders- hypersensitivity reactions, autoimmunity, immunodeficiency states etc.
- Genetic/metabolic disorders- examples gout, diabetes mellitus etc.

Nomenclature:

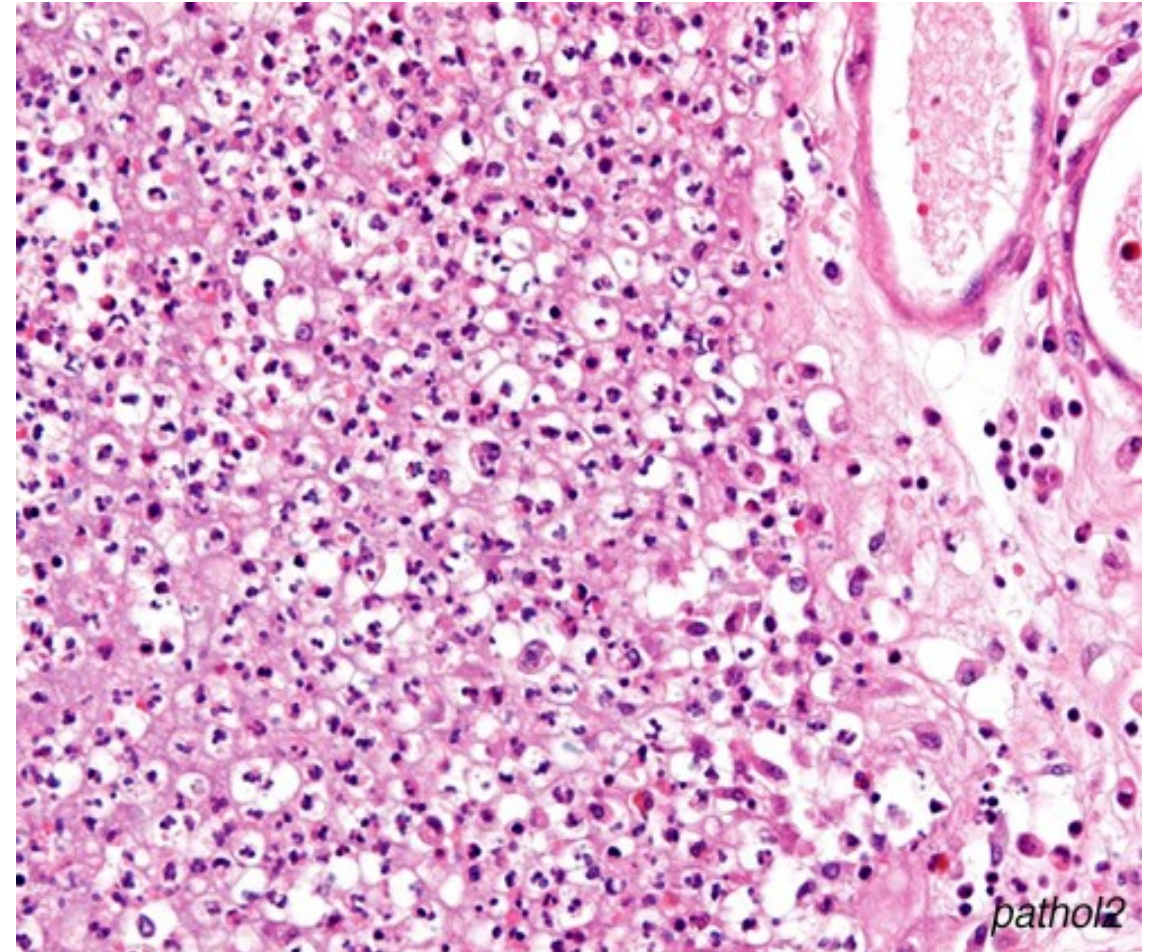
- The nomenclatures of inflammatory lesion are usually indicated by the **suffix 'itis'**.
- Thus, inflammation of the appendix is called appendicitis and that of meninges as meningitis, conjunctivitis, etc.
- However, like any rule, it has its own exceptions examples pneumonia, etc....

Classification

- Inflammation is classified in to two based on duration of the lesion and histologic appearances into acute and chronic inflammation.

Acute inflammation

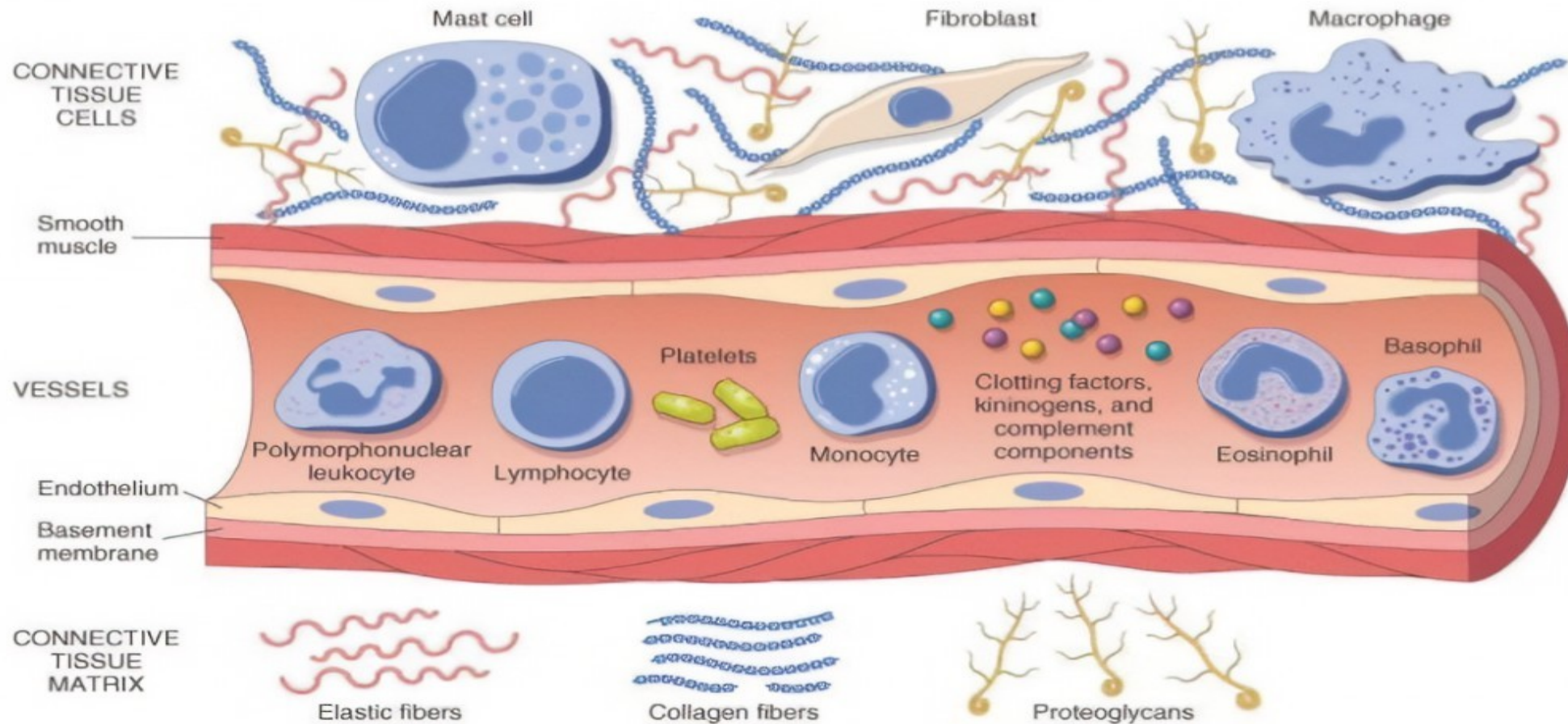
- Acute inflammation is an immediate and early response to an injurious agent and it is relatively of short duration, lasting for minutes, several hours or few days.
- It is characterized by **exudation of fluids** and plasma proteins and the emigration of predominantly **neutrophilic** leucocytes to the site of injury



mediators

- Toll like receptors.
- Arachidonic acid
 - COX- Produces Prostaglandins
 - PGD2,PGE2,PGI2- mediated vasodilation and vascular permeability
 - PGE2 mediates pain
 - **Antipain medication**
 - LOX- Leukotrien production- Attract and activate Neutrophil
- Mast cells
- Complement

Components of Inflammation



Events of acute inflammation:

1. Vascular events

Exit of cells and plasma cells from the blood into tissue requires alteration of the blood vessels

- ✓ Transient vasoconstriction (lasting only for seconds)
- ✓ **Vasodilation**.
 - ✓ Histamine,
- ✓ **Increase vascular permeability**
 - ✓ Contraction of endothelial cells and/or direct endothelial injury
 - ✓ Histamine, bradykinin, leukotriens

2. Cellular events

- The cellular response has the following stages:

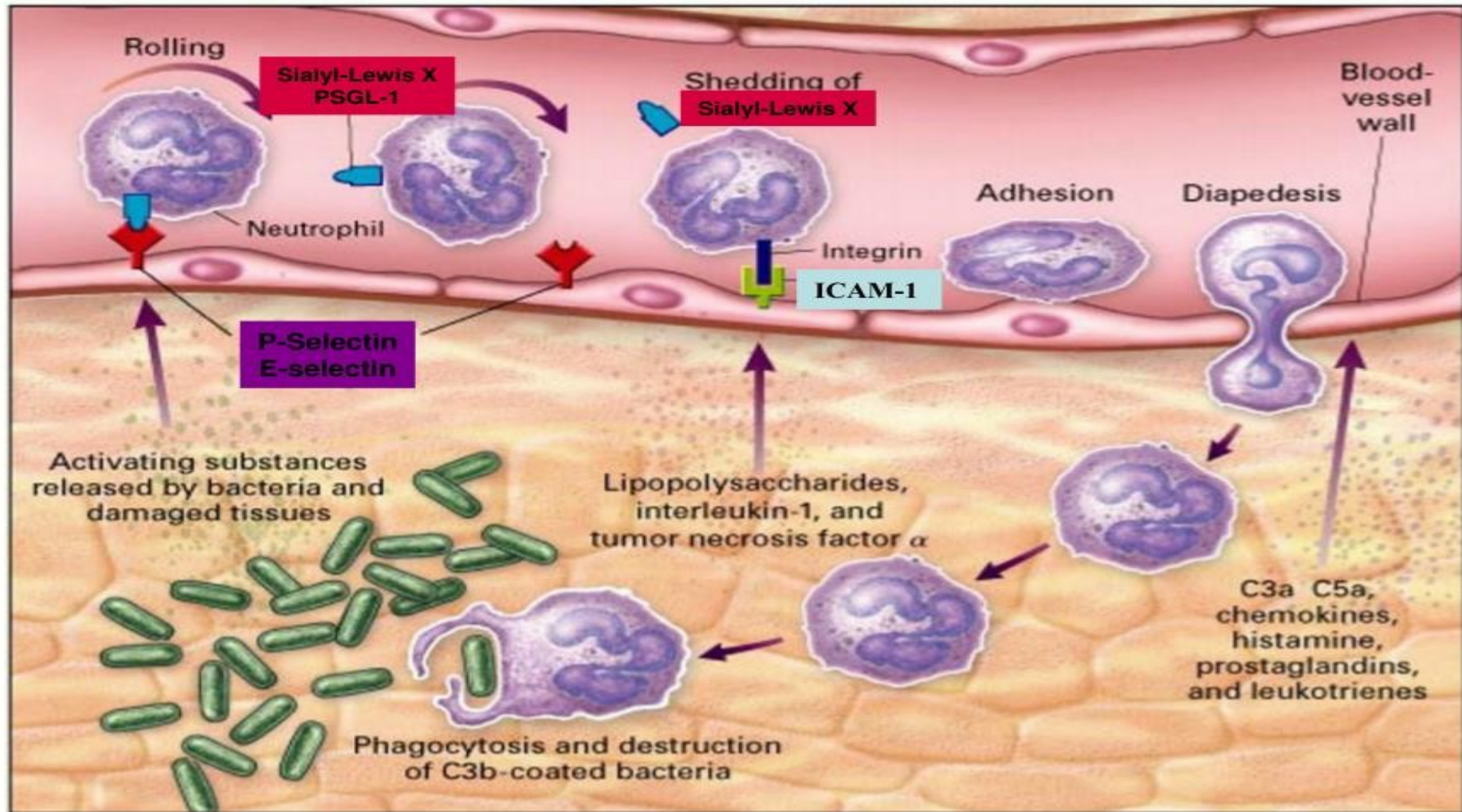
➤ **Leucocyte recruitment:**

- a. Margination:** is a peripheral positioning of white cells along the endothelial cells.
- b. Rolling:** tumbling of leukocytes slowly along the endothelium
 - In time, the endothelium can be virtually lined by white cells.
 - Selectin- speed bumps
- c. Adhesion:** the binding of leukocytes with endothelial cells.
 - Cellular adhesion molecules (endothelium), Integrins upregulated on leukocyte

d. Transmigration (Diapedesis)

- Leukocytes escape from venules and small veins but only occasionally from capillaries.
- The movement of leukocytes is via extending pseudopodia through the vascular wall
- The most important mechanism of leukocyte emigration is via widening of interendothelial junctions after endothelial cells contractions.
- The basement membrane is disrupted and resealed thereafter immediately

Neutrophil Recruitment: the Multistep Model



e. Chemotaxis:

- A unidirectional attraction of leukocytes from vascular channels towards the site of inflammation within the tissue space guided by chemical gradients.
- The most important chemotactic factors for neutrophils are components of the complement system (C5a), bacterial and mitochondrial products of arachidonic acid metabolism such as leukotriene B₄ and cytokines (IL-8).
- All granulocytes, monocytes and to lesser extent lymphocytes respond to chemotactic stimuli.

➤ Phagocytosis

- Phagocytosis is the process of engulfment and internalization by specialized cells of particulate material, which includes invading microorganisms, damaged cells, and tissue debris.
- These phagocytic cells include polymorphonuclear leukocytes (particularly neutrophils), monocytes and tissue macrophages.
- Phagocytosis involves three distinct but interrelated steps

1. Recognition and attachment

2. Engulfment

3. Killing or degradation

- The ultimate step in phagocytosis of bacteria is killing and degradation.
- There are two forms of bacterial killing

a). Oxygen-independent mechanism:

- This is mediated by some of the constituents of the primary and secondary granules of polymorphonuclear leukocytes.
- ✓ E.g. Bactericidal permeability increasing protein (BPI),
Lysozymes,

Lactoferrin, Major basic protein Defenses

- It is probable that bacterial killing by lysosomal enzymes is inefficient and relatively unimportant compared with the oxygen dependent mechanisms.
- The lysosomal enzymes are, however, essential for the degradation of dead organisms within phagosomes.

b) Oxygen-dependent mechanism:

- More effective compared to oxygen independent killing
- The end product is hypochlorous acid

Cardinal signs of acute inflammation

- **Redness** (rubor): which is due to dilation of small blood vessels
- **Heat** (calor): which results from increased blood flow (hyperemia) due to regional vascular dilation
- **Swelling** (tumor): which is due to accumulation of fluid in the extravascular space which, in turn, is due to increased vascular permeability.
- **Pain** (dolor): which partly results from the stretching & destruction of tissues due to inflammatory edema and in part from pus under pressure in an abscess cavity.
- **Loss of function**: The inflamed area is inhibited by pain while severe swelling may also physically immobilize the tissue.

Redness



Swelling



Function loss



➤ **Effects of acute inflammation**

A. Beneficial effects

- Dilution of toxins
- Protective antibodies
- Fibrin formation
- Plasma mediator systems provisions
- Cell nutrition
- Promotion of immunity

B. Harmful effects

- Tissue destruction
- Swelling
- Inappropriate response

Systemic manifestations

- Fever
 - Bacterial products stimulates leukocytes to release cytokines such as IL1 and TNF that increase COX-
 - PGE2 stimulates hypothalamus and induce fever
 - NSAID- aspirin inhibit PG synthesis
- Leukocytosis
- Acute phase reactant

➤ **Four Outcomes of Acute Inflammation**

- Complete resolution with regeneration
- Complete resolution with scarring
- Abscess formation
- Transition to chronic inflammation

CHRONIC INFLAMMATION

Definition: Chronic inflammation can be defined as a prolonged inflammatory process (weeks or months) where an active inflammation, tissue destruction and attempts at repair are proceeding simultaneously.

➤ **Causes of Chronic Inflammation**

- Following bout of acute inflammation
- Persistent infections
- Infections with certain organisms, including viral infections, mycobacteria, parasitic infections, and fungal infections
- Autoimmune disease
- Response to foreign material
- Response to malignancy

➤ **Cells of chronic inflammation**

- **Monocytes and Macrophages**

- Liver : Kupffer cells

- Reticuloendothelial system : sinus histiocytes

- CNS : microglial cells

- Lungs : alveolar macrophages

- skin : Langerhans cells

- ❖ **Macrophages are scavenger cells of the body**

Other cells in chronic inflammation

- T-Lymphocytes
- B-lymphocytes and Plasma cells
- Mast cells and eosinophils
- **N.B** :Though neutrophils are hallmarks of acute inflammatory reactions, large numbers of neutrophils may be seen in some forms of chronic inflammation, notably chronic osteomyelitis, actinomycosis, & chronic lung diseases induced by smoking and other stimuli.

➤ Granulomatous Inflammation:

- Granulomatous inflammation is a distinctive pattern of chronic inflammation characterized by **aggregates of activated macrophages** that assume an **epithelioid appearance**.

➤ **Composition of a granuloma is as follows:**

- **Epithelioid cells:** located centrally, form when IFN- γ transforms

macrophages to epithelioid cells.

-They are enlarged cells with abundant pink cytoplasm.

- **Multinucleated giant cells:** located centrally, are formed by the fusion of epithelioid cells.

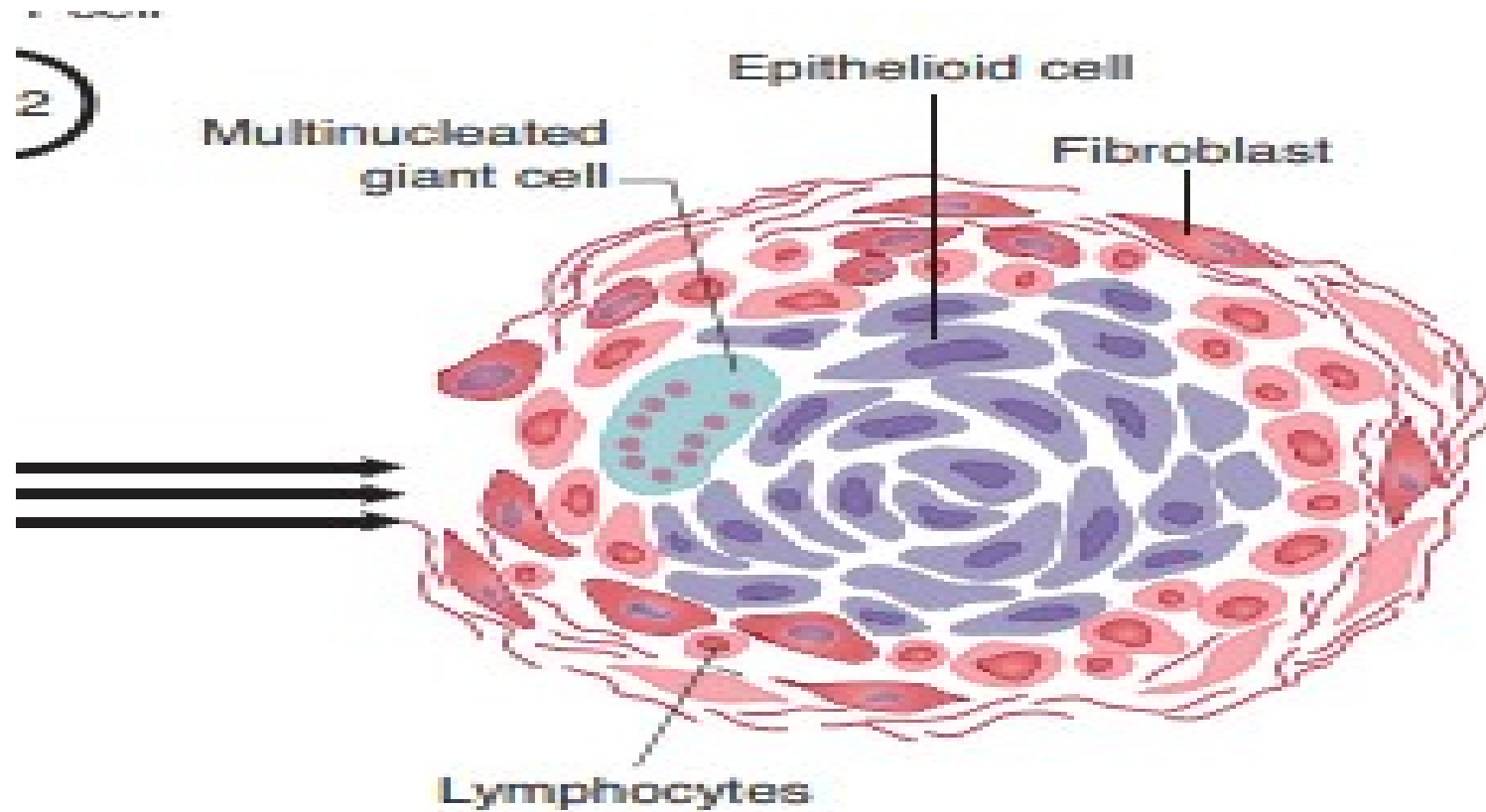
1. *Langhans-type giant cell* (peripheral arrangement of nuclei)

2. *Foreign body type giant cell* (haphazard arrangement of nuclei)

- **Lymphocytes and plasma cells:** are present at the periphery

- **Central necrosis:** occurs in granulomata due to excessive enzymatic break down and is commonly seen in **Mycobacterium tuberculosis** infection as well as fungal infections and a few bacterial infections. Because of the public health risk of tuberculosis, necrotizing granulomas should be considered tuberculosis until proven otherwise.

Granuloma



- Granulomatous diseases include :tuberculosis (caseating granulomas), cat scratch fever, syphilis, leprosy, fungal infections (e.g., coccidioidomycosis), parasitic infections (e.g., schistosomiasis), foreign bodies, beryllium, and sarcoidosis.

Acute vs chronic inflammation

<u>Characteristics</u>	<u>Acute inflammation</u>	<u>Chronic inflammation</u>
Duration	Short	Relatively long
Pattern	Stereotyped	Varied
Predominant cell	Neutrophils	Macrophages,
	Lymphocytes	plasma cells
Tissue destruction	Mild to moderate	Marked
Fibrosis	Absent	Present

Healing

Outline

- Cell cycle
- Repair by regeneration
- Repair by fibrosis
- Skin wound healing
- Complications of wound healing

HHealing

- Healing is initiated when inflammation begins.
- Occurs via a combination of **regeneration** and **repair**.
- Tissue repair involves 5 overlapping processes:
 - ✓ Hemostasis (coagulation, platelets)
 - ✓ Inflammation (neutrophils, macrophages, lymphocytes, mast cells)
 - ✓ Regeneration (stem cells and differentiated cells)
 - ✓ Fibrosis (macrophages, granulation tissue [fibroblasts, angiogenesis], type III collagen)
 - ✓ Remodeling (macrophages, fibroblasts, converting collagen III to I)

REGENERATION

- Replacement of damaged tissue with native tissue; dependent on regenerative capacity of tissue
- proliferation of residual cells to restore damaged cells.

Labile tissues possess stem cells that continuously cycle to regenerate the tissue.

1. Small and large bowel (stem cells in mucosal crypts)
2. Skin (stem cells in basal layer,)
3. Bone marrow (hematopoietic stem cells)

Stable cells

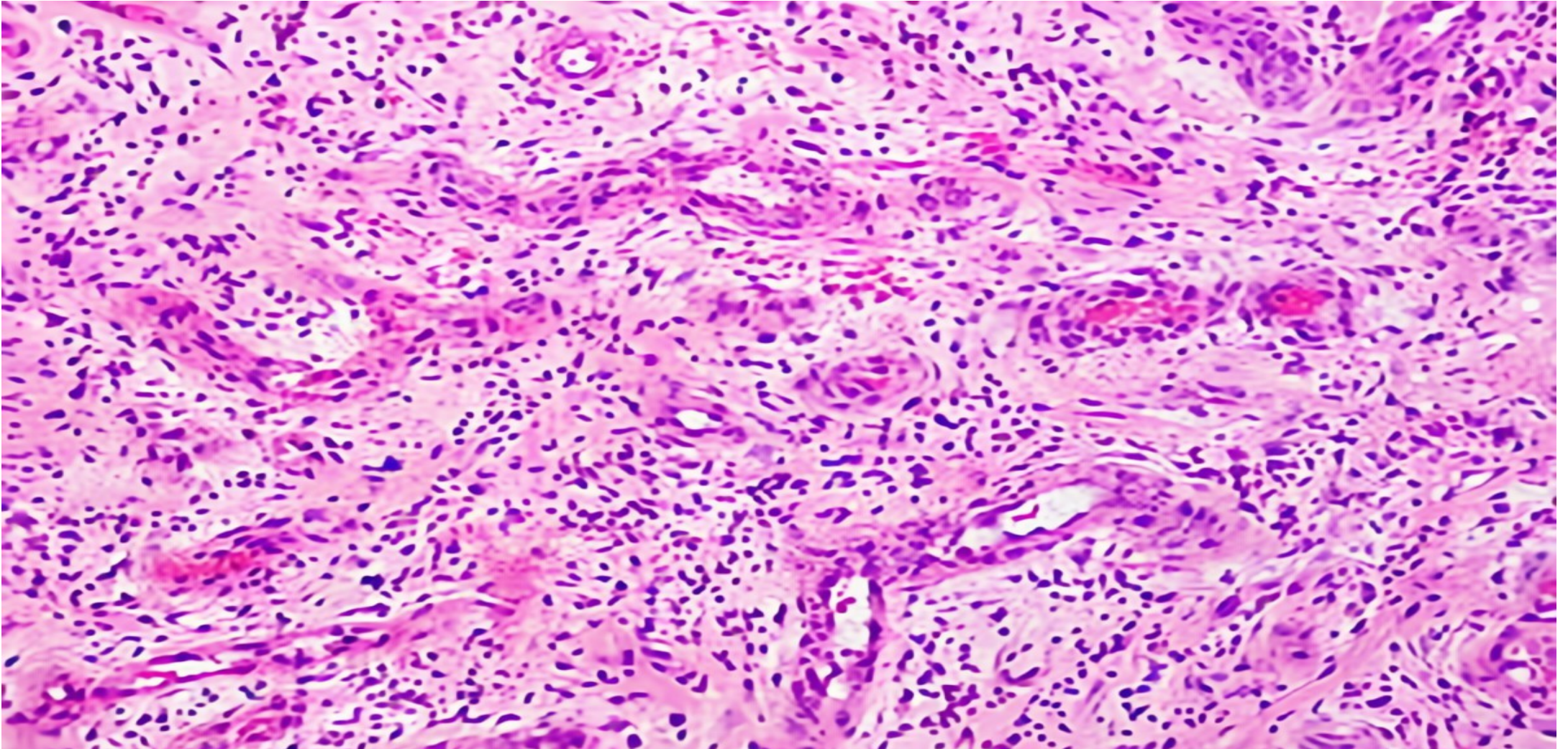
- Cells of these tissues are quiescent and have only minimal proliferative activity in their normal state.
- They are capable of dividing in response to injury or loss of tissue mass.
- Stable cells constitute the parenchyma of most solid tissues, such as liver, kidney, and pancreas.
- They also include endothelial cells, fibroblasts, and smooth muscle cells;
- With the exception of liver, stable tissues have a limited capacity to regenerate after injury

- **Permanent cells** (few stem cells and/or differentiated cells with the capacity to replicate) have a very low level of replicative capacity.
- Examples include neurons, skeletal muscle cells and cardiac muscle

2. REPAIR BY FIBROSIS

- Replacement of damaged tissue with fibrous scar
- Occurs when regenerative stem cells are lost (e.g., deep skin cut) or when a tissue lacks regenerative capacity (e.g., healing after a myocardial infarction)
- **Granulation tissue formation is the initial phase of repair .**
 - Consists of fibroblasts (deposit type III collagen), capillaries (provide nutrients), and myofibroblasts (contract wound)
- **Eventually results in scar formation**, in which type III collagen is replaced with type I collagen

Granulation tissue



1. Type III collagen is pliable and present in granulation tissue, embryonic tissue, uterus, and keloids.
2. Type I collagen has high tensile strength and is present in skin, bone, tendons, and most organs.
3. Collagenase removes type III collagen and requires zinc as a cofactor

Types of Wound Healing (Skin)

- **Primary union** (healing by first intention) occurs when wounds are closed physically with sutures, metal staples, dermal adhesive, etc.
 - ❑ *N.B Leads to minimal scar formation*
- **Secondary union** (healing by secondary intention) occurs when wounds are allowed to heal by wound contraction and is mediated by *myofibroblasts* at the edge of the wound
- E.g. When cell or tissue loss is more extensive, such as in large wounds, abscesses, ulceration, and ischemic necrosis (infarction) in parenchymal organs

Aberrations or complications in Wound Healing

- **Delayed wound healing** may be seen in wounds complicated by foreign bodies, infection, ischemia, diabetes, malnutrition, scurvy, etc.
 1. Infection (most common cause; *S aureus* is the most common offender)
 2. Vitamin C, copper, or zinc deficiency
 3. Other causes include foreign body, ischemia, diabetes, and malnutrition, location of wound, mechanical pressure

- **Hypertrophic scar** results in a prominent scar that is localized to the wound, due to excess production of granulation tissue and collagen. It is common in burn patients.



Keloid formation is a genetic predisposition that is common in African Americans.

- It tends to affect the earlobes, face, neck, sternum, and forearms, and it may produce large tumor-like scars **extending beyond the injury site**.
- There is **excess production of collagen** that is predominantly type III.
- Genetic predisposition (more common in African Americans)
- Classically affects earlobes, face, and upper extremities



➤ **Contracture**

- Contraction in the size of a wound is an important part of the normal healing process.
- An exaggeration of this process gives rise to **contracture** and results in deformities of the wound and the surrounding tissues.
- Contractures are particularly prone to develop on the palms, the soles, and the anterior aspect of the thorax.
- Contractures are commonly seen after serious burns and can compromise the movement of joints



References

- Robbins and cotran pathologic basics of disease 10th edition

Thank you!